> Question 3: Automate application deployment using Jenkins by configuring build triggers, post-build actions, and deployment steps.
>
> Marks: 5 | Course outcome: CO1 | Environment: Ubuntu 24.04 LTS (WSL2), Git 2.43.0, Jenkins 2.580.2 LTS

Deployment is fully automated: Jenkins polls GitHub for new commits, builds and tests the application, deploys it to the local deployment directory, and verifies the running service - all inside the same pipeline run.

1. Build trigger configuration

job configuration

```
Build Triggers -> Poll SCM
Schedule: H/5 * * * *   (every 5 minutes, hash-distributed minute)
```

Because the WSL2 lab environment is not reachable from the public internet, GitHub cannot deliver webhooks; SCM polling achieves the same automation by having Jenkins check GitHub for new commits on a schedule.

2. Deployment steps (deploy stage + deploy.sh)

deploy/deploy.sh (committed to the repository)

```
#!/bin/bash
# Deployment script invoked by the Jenkins deploy stage.
set -e
APP_DIR="${DEPLOY_DIR:-$HOME/enp461-deploy}"
PYTHON="${VENV:+$VENV/bin/python3}"
PYTHON="${PYTHON:-python3}"

mkdir -p "$APP_DIR"
rsync -a --exclude ".git" --exclude "__pycache__" --exclude ".pytest_cache" --exclude \
      ".venv" ./ "$APP_DIR/"

# stop any previous instance owned by this user
pkill -f "python3 app.py" 2>/dev/null || true

cd "$APP_DIR/app"
JENKINS_NODE_COOKIE=dontKillMe nohup "$PYTHON" app.py > "$APP_DIR/deploy.log" 2>&1 &

# verify the health endpoint (up to 10 attempts)
for i in $(seq 1 10); do
  sleep 2
  if curl -sf http://localhost:5000/health > /dev/null; then
    echo "Deployment verified: health OK (attempt $i)"
    curl -s http://localhost:5000/
    exit 0
  fi
done
echo "Deployment FAILED: health endpoint not responding" >&2
cat "$APP_DIR/deploy.log" >&2
exit 1
```

3. Post-build actions

Jenkinsfile post section

```
post {
    success { echo "Pipeline SUCCESS: application built, tested and deployed." }
    failure { echo "Pipeline FAILED - check the stage logs." }
    always { echo "Cleaning workspace temporary files..."; cleanWs() }
}
```

build #2 console output (post actions)

```
[Pipeline] echo
Pipeline SUCCESS: application built, tested and deployed.
[WS-CLEANUP] Deleting project workspace...
[WS-CLEANUP] done
Finished: SUCCESS
```

4. Proof of the automated trigger

git-polling-log.txt - the poll detects commit eaa01d3 (Changes found)

```
Started on Nov 4, 2026, 6:14:33 PM
Using strategy: Default
[poll] Last Built Revision: Revision eaa01d3d856725c9504bf648a2f3704e02fe68b6 (refs/remotes/origin/main)
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential github-creds
 > git --version # timeout=10
 > git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials GitHub PAT for Meenakshi (ENSP461 Lab 1)
 > git ls-remote -h -- https://github.com/meenakshi16rp/Devops-lab-1-.git # timeout=10
Found 2 remote heads on https://github.com/meenakshi16rp/Devops-lab-1-.git
[poll] Latest remote head revision on refs/heads/main is: eaa01d3d856725c9504bf648a2f3704e02fe68b6 - \
      already built by 3
Using strategy: Default
[poll] Last Built Revision: Revision eaa01d3d856725c9504bf648a2f3704e02fe68b6 (refs/remotes/origin/main)
Selected Git installation does not exist. Using Default
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 > git ls-remote -h -- https://github.com/meenakshi16rp/Devops-lab-1-.git # timeout=10
Found 2 remote heads on https://github.com/meenakshi16rp/Devops-lab-1-.git
[poll] Latest remote head revision on refs/heads/main is: eaa01d3d856725c9504bf648a2f3704e02fe68b6 - \
      already built by 3
Done. Took 1.8 sec
No changes
```



Figure: Git polling log in Jenkins

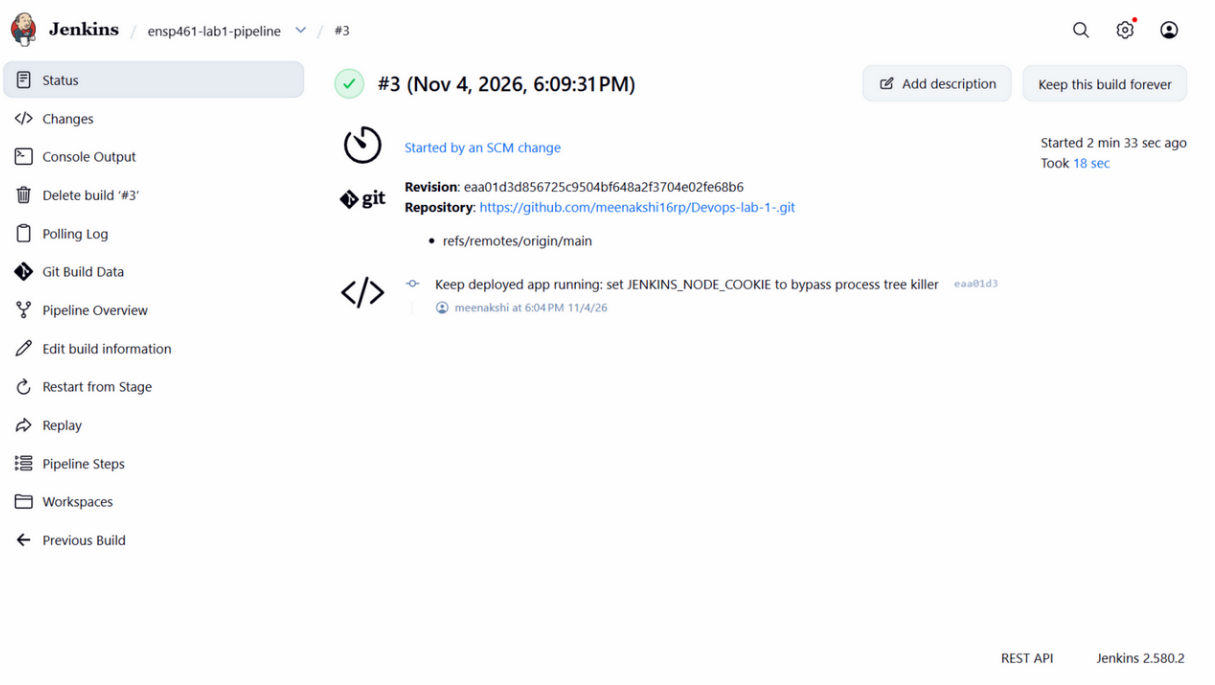


Figure: Build #3 - cause: 'Started by an SCM change', result SUCCESS

5. Deployed application verification

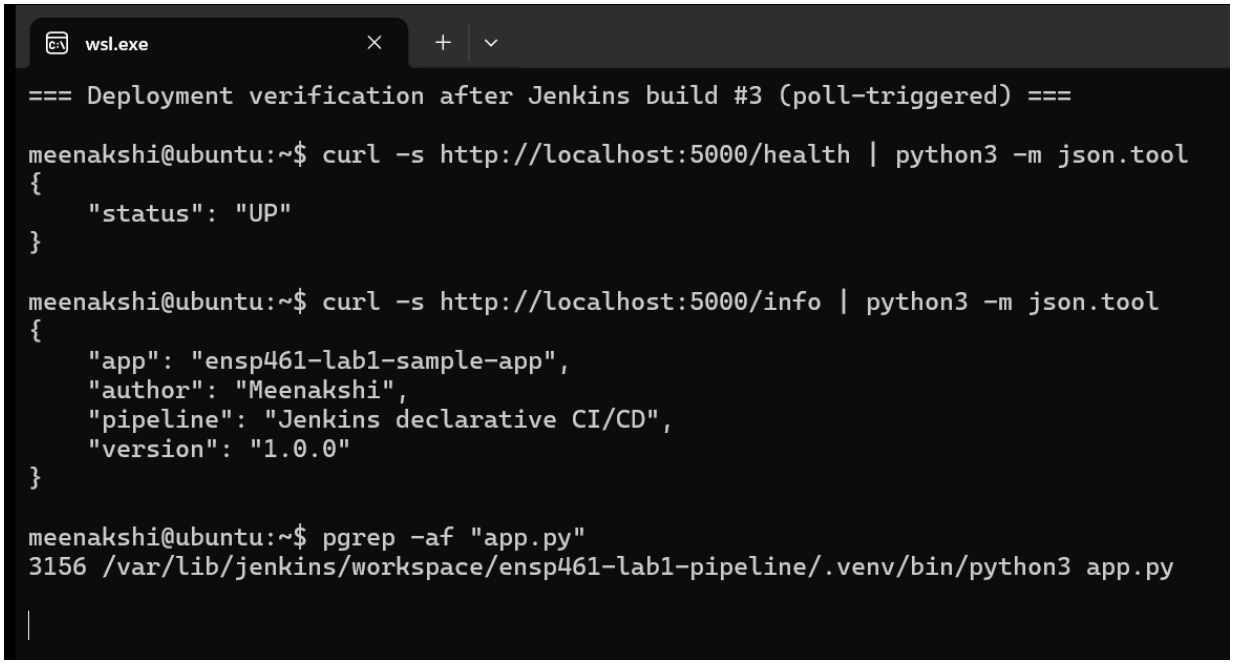


Figure: The app deployed by build #3 answers on http://localhost:5000

health and info endpoints after the automated build

```
curl -s http://localhost:5000/health
{"status":"UP"}
curl -s http://localhost:5000/info
{"app":"ensp461-lab1-sample-app","author":"Meenakshi","pipeline":"Jenkins declarative \
      CI/CD","version":"1.0.0"}
```

6. Observations

- The first automated attempt (build #1) failed because the deploy script used the system Python without Flask; reusing the pipeline virtualenv fixed it - the fix itself went through the same automated pipeline.

- Jenkins kills child processes at the end of a build; JENKINS_NODE_COOKIE=dontKillMe keeps the deployed service alive after the run.
- The retry loop around the health check makes the Deploy stage robust against slow startups.